

1.3 Process $C \rightarrow D$

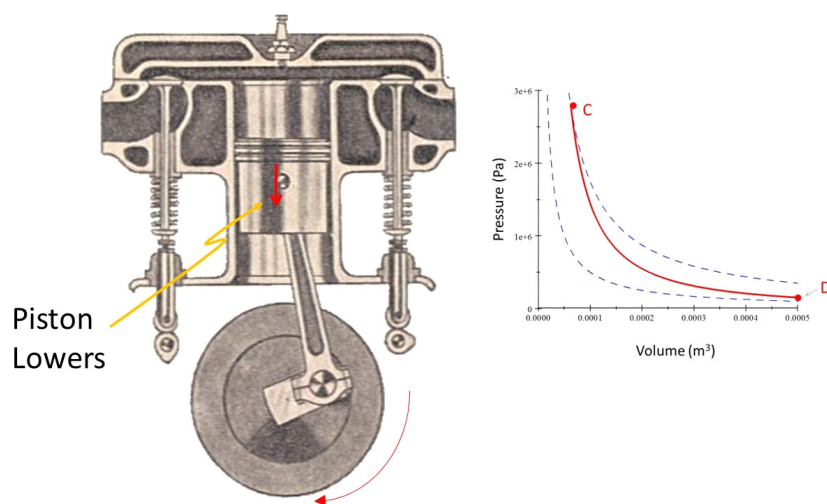
Assignment:

Fill in the table for Process $C \rightarrow D$

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
D				2.0525×10^{-2}		—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
CD						

ANSWER:

Process CD (Power Stroke)



Process $C \rightarrow D$ is another adiabatic process, this time an expansion. This is sometimes called the “power stroke” because the hot, high pressure gas pushes the piston down. This push is what makes the car or lawn mower (snow blower) go. We know that $Q_{CD} = 0$ so $\Delta E_{CD} = w_{CD}$

To find the ideal gas state variables at D let's start with

$$P_i V_i^\gamma = P_f V_f^\gamma$$

again, then

$$P_D = P_C \frac{V_C^\gamma}{V_D^\gamma}$$

but from our PV diagram we see $V_C = V_B$ and $V_D = V_A$. This makes sense because the cylinder and piston system maximum and minimum volumes don't change. So we can write

$$P_D = P_C \frac{V_B^\gamma}{V_A^\gamma} = 8^{-\gamma} P_C$$

and knowing this we can find the temperature at D using the ideal gas law

$$T_D = \frac{P_D V_D}{nR}$$

Let's numerically calculate P_D and T_D at this point.

$$\begin{aligned} P_D &= 8^{-1.4} (2793.1) \\ &= 151.97 \text{ kPa} \end{aligned}$$

and

$$\begin{aligned} T_D &= \frac{(151.97 \text{ kPa}) (0.0005 \text{ m}^3)}{(2.0525 \times 10^{-2} \text{ mol}) (8.314 \frac{\text{J}}{\text{mol K}})} \\ &= 445.28 \text{ K} \end{aligned}$$

And while we are at it we can complete our state D by calculating E_D

$$\begin{aligned} E_D &= \frac{5}{2} n R T_D \\ &= \frac{5}{2} (2.0525 \times 10^{-2} \text{ mol}) \left(8.314 \frac{\text{J}}{\text{mol K}} \right) (445.28 \text{ K}) \\ &= 189.96 \text{ J} \end{aligned}$$

Now for the work done, we again use our new adiabatic equation

$$\begin{aligned} w_{CD} &= \left(\frac{1}{\gamma - 1} \right) (P_D V_D - P_C V_C) \\ &= \left(\frac{1}{\gamma - 1} \right) (8^{-\gamma} P_C V_A - P_C V_B) \\ &= \left(\frac{P_C}{\gamma - 1} \right) (8^{-\gamma} V_A - V_B) \end{aligned}$$

all of which we know, so

$$\begin{aligned} W_{CD} &= \left(\frac{2793.1 \text{ kPa}}{1.4 - 1} \right) (8^{-1.4} (0.0005 \text{ m}^3) - 6.25 \times 10^{-5} \text{ m}^3) \\ &= -246.46 \text{ J} \end{aligned}$$

which completes our set for process CD

State	T (K)	P (kPa)	V (m ³)	n (mol)	E _{int} (J)	—
C	1023	2793.1	6.25×10^{-5}	2.0525×10^{-2}	436.42	—
D	445.28	151.97	0.0005	2.0525×10^{-2}	189.96 J	—
Process	Q (J)	w _{int} (J)	ΔE _{int} (J)	w _{eng} (J)	Q _h	Q _c
CD	0	-246.46	-246.46	246.46	0	0